

SWINBURNE UNIVERSITY OF TECHNOLOGY

ZERO-COST OPTION

Unit Code: Computing Technology Project A (COS40005)
Project: Internal Assistant for Enterprise Knowledge (Slack chatbot with RAG)
Client: Katalon, Inc. (replacing the original VIB client)
Team: Group 7
Prepared by: Truong Ngoc Gia Hieu (Project Manager)
Version: v1.0 (Draft for discussion)
Date: 11TH Nov, 2026

Table of Contents

A. Purpose and Scope....3
B. Environment at a Glance....3
C. Components in Detail....5
3.1 Okta Integrator Free Plan (identity provider)....5
3.2 Slack Developer Sandbox and the Slack app....5
3.3 Confluence Free plan and 14-day Standard trial....5
3.4 AWS Lambda with a function URL....6
3.5 Amazon DynamoDB with Time to Live....6
3.6 Amazon Bedrock Knowledge Bases with Titan Text Embeddings V2....6
3.7 Amazon S3 Vectors (vector store)....6
3.8 Amazon Nova Lite on Bedrock (answer generation)....7
3.9 AWS Systems Manager Parameter Store (standard tier)....7
3.10 EventBridge Scheduler, CloudWatch and AWS Budgets....7
D. Workflows and Sequence Diagrams....8
4.1 Provisioning test personas and signing in....8
4.2 Answering a question with two-layer access control....8
4.3 Keeping the knowledge base fresh....10
E. Testing in This Environment....11
5.1 Test personas....11
5.2 How access control is tested....11
F. Timeline....11
G. Cost Summary....12
H. Risks and Mitigations....12
I. Alignment with Evaluation Criteria....13
References....14

A. Purpose and Scope

This document describes in detail the **Option 1: Zero-Cost Environment** for the Katalon internal-knowledge Slack assistant built by Group 7. It explains every technology in the environment, why it was chosen, how it helps the project, and how the parts work together, using an architecture diagram and three sequence diagrams. It complements the *Environment Options* overview (Group 7, 2026a) and the matching project budget workbook (Group 7, 2026b).

The environment keeps the architecture requested by the client: Confluence is the only knowledge source, Slack is the only user interface, and Amazon Bedrock provides retrieval-augmented generation (RAG). It is a **test (staging) environment**. Katalon's production Slack, Confluence and identity provider are not used, because an external student team is unlikely to be given access to them. All prices were checked on 11 November 2026, are in US dollars for the US East (N. Virginia) Region, and may change.

B. Environment at a Glance

Table 1 summarises the environment. Figure 1 shows how the components are connected.

Layer	Technology	Role in the project	Cost to the team
Identity provider	Okta Integrator Free Plan	Single sign-on (SAML) and SCIM provisioning for test personas	US$0
Chat interface	Slack Developer Sandbox + Slack app	Where employees ask questions and receive answers	US$0
Knowledge source	Confluence Free, then 14-day Standard trial	Sample spaces that copy Katalon's structure, with real permissions	US$0
Compute	AWS Lambda (bot and sync functions) with a function URL	Runs all logic without servers	Always-free allowance
Caches and state	Amazon DynamoDB with Time to Live (TTL)	Identity map and permission cache	Always-free allowance
Retrieval	Amazon Bedrock Knowledge Bases + Titan Text Embeddings V2	Chunking, embedding and filtered retrieval	Paid from AWS credits
Vector store	Amazon S3 Vectors	Stores embeddings with page-level metadata	Paid from AWS credits
Generation	Amazon Nova Lite on Bedrock (Nova 2 Lite if accuracy needs it)	Writes grounded answers from permitted chunks	Paid from AWS credits
Secrets	AWS Systems Manager Parameter Store (standard tier)	Holds the Slack and Confluence tokens	US$0
Scheduling and monitoring	Amazon EventBridge Scheduler, Amazon CloudWatch, AWS Budgets	Runs the sync, records logs and freshness, alerts on spending	Free allowances

Table 1. Components of the environment

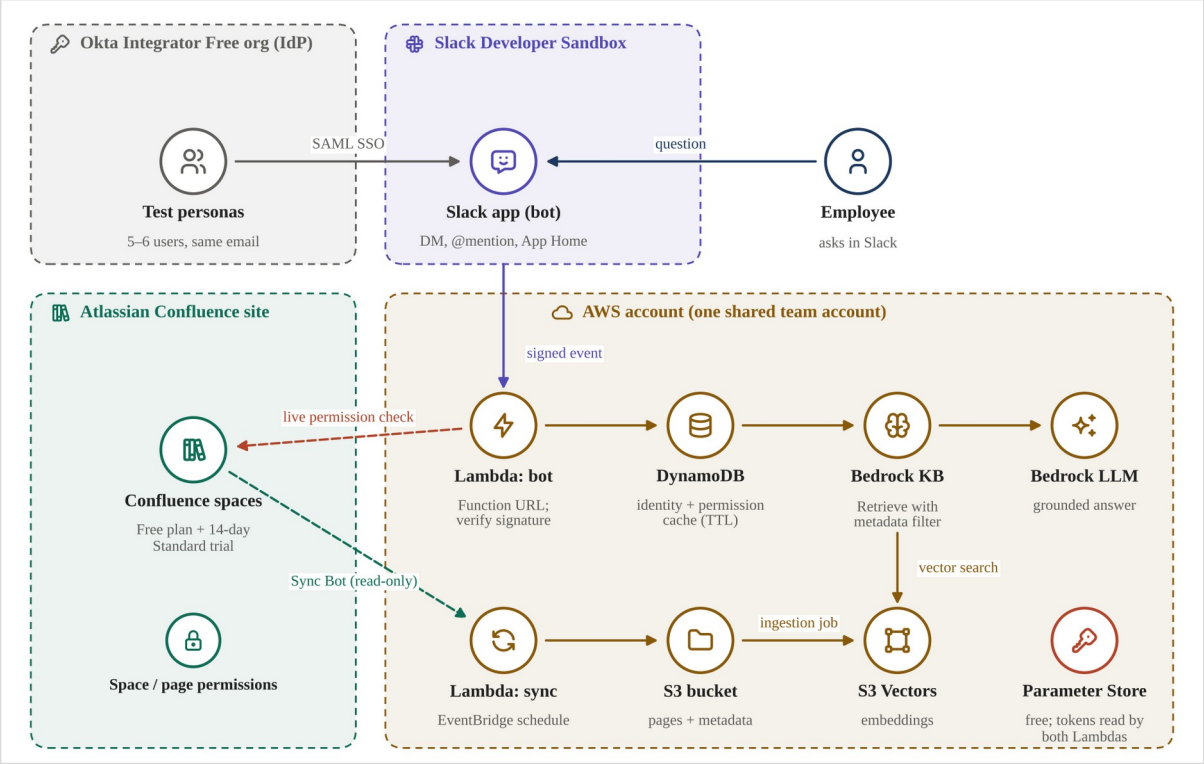


Figure 1. Architecture of Option 1: icons show each service; dashed arrows are permission traffic

C. Components in Detail

Each component below is described with three questions: **what it is**, **why the team uses it**, and **how it helps the project**. The evaluation weights referred to are the client's: accuracy 40%, cost and maintainability 30%, security 20% and creativity 10%.

3.1 Okta Integrator Free Plan (identity provider)

Question	Answer
What it is	A free Okta organisation intended for building and testing integrations. It supports SAML and OIDC single sign-on and SCIM provisioning for custom app integrations (Okta, n.d.).
Why we use it	The Slack sandbox is an Enterprise Grid environment, and Grid requires all users to sign in with SSO, so an identity provider is mandatory (Slack, n.d.-h). Okta offers this at no cost.
How it helps the project	The sign-in path becomes the same as at Katalon: one company login, then Slack. SCIM creates the Slack test users with the same email addresses that the bot later uses to match people to Confluence accounts.

3.2 Slack Developer Sandbox and the Slack app

Question	Answer
What it is	A free, isolated Enterprise Grid sandbox from the Slack Developer Program, limited to eight users and two guests and active for six months by default (Slack, n.d.-a). The bot is a Slack app installed with a bot token (`xoxb`) and the scopes `chat:write`, `app_mentions:read`, `im:history`, `users:read` and `users:read.email`.
Why we use it	It gives the team a real enterprise Slack without touching Katalon's workspace. Eight users are enough for the test personas, and the two guest seats let the team prove that guests are refused.
How it helps the project	Every event carries the sender's Slack ID. The bot calls `users.info` to read the email address and the guest flags `is_restricted` and `is_ultra_restricted` (Slack, n.d.-c, n.d.-d, n.d.-e). Requests are authenticated with the signing secret: Slack signs each request with HMAC-SHA256, and the app compares the signature and rejects timestamps older than five minutes (Slack, n.d.-f).

3.3 Confluence Free plan and 14-day Standard trial

Question	Answer
What it is	A Confluence Cloud site on the Free plan (up to 10 users) for most of the project, upgraded to a 14-day Standard trial before integration testing and UAT. At the end of a trial without a card, the site returns to the Free plan automatically (Atlassian, n.d.-a).
Why we use it	The Free plan does not allow space or page permission settings (Atlassian, n.d.-c), so it is used only for work that does not need permissions. The trial supplies real permissions at the moment they are needed, at no cost.
How it helps the project	The site holds sample spaces that copy Katalon's structure (Intranet, Deal Desk, Partner Program and True Platform) and the group names the client shared. A read-only **Sync Bot** account with an API token reads pages and checks permissions; it never edits content (Atlassian, n.d.-e).

3.4 AWS Lambda with a function URL

Question	Answer
What it is	Serverless functions: a **bot function** that handles Slack events and a **sync function** that keeps the knowledge base current. The bot function is reached through a Lambda function URL, a dedicated HTTPS endpoint (Amazon Web Services [AWS], n.d.-f).
Why we use it	There are no servers to patch, and Lambda's free tier includes one million requests per month (AWS, n.d.-g). A function URL removes the need for Amazon API Gateway, whose free calls are not part of the AWS Free plan (AWS Builder Center, 2026).
How it helps the project	Slack expects an HTTP 2xx response within three seconds (Slack, n.d.-g). The bot function verifies the signature, acknowledges at once and continues the work asynchronously, which keeps the user experience responsive.

3.5 Amazon DynamoDB with Time to Live

Question	Answer
What it is	A serverless key-value database holding two caches: the **identity map** (Slack ID to email to Atlassian account ID) and the **permission cache** (allowed page IDs for each user).
Why we use it	Calling Slack and Confluence for every question would be slow and could hit rate limits. TTL deletes expired items automatically and without consuming write throughput (AWS, n.d.-h), and the free tier covers 25 GB of storage (AWS, n.d.-i).
How it helps the project	Answers are faster and cheaper (cost criterion). A 15-minute TTL means that permissions removed in Confluence stop being used quickly, and the live check in Section D removes even that delay for the pages actually used.

3.6 Amazon Bedrock Knowledge Bases with Titan Text Embeddings V2

Question	Answer
What it is	A managed RAG service that reads documents from an S3 bucket, splits them into chunks, embeds them and answers `Retrieve` calls. Titan Text Embeddings V2 produces 1,024-dimension vectors and accepts up to 8,192 tokens per input (AWS, n.d.-j).
Why we use it	It removes the need to write chunking and embedding code. Each page is stored with a `.metadata.json` file containing its page ID, space and groups, and `Retrieve` can filter on these attributes (AWS, 2024, n.d.-k).
How it helps the project	Retrieval only searches pages the asker may read (security), and only changed pages are re-processed during a sync (freshness and cost). Embedding costs about US$0.02 per million tokens (AWS, n.d.-a).

3.7 Amazon S3 Vectors (vector store)

Question	Answer
What it is	Vector storage built into Amazon S3, generally available since December 2025 and supported as a Bedrock knowledge-base vector store (AWS, 2025b).
Why we use it	The default quick-create store, Amazon OpenSearch Serverless, keeps a minimum of four capacity units running and costs roughly US$700 per month even with no queries (Mallesh, n.d.). S3 Vectors has no minimum charge and costs nothing while idle (amitrix, n.d.).
How it helps the project	For a corpus of a few hundred pages, storage and queries cost cents per month, which protects the AWS credits (AWS, n.d.-d). This is the single most important cost decision in the environment.

3.8 Amazon Nova Lite on Bedrock (answer generation)

Question	Answer
What it is	A low-cost foundation model on Bedrock, priced at about US$0.06 per million input tokens and US$0.24 per million output tokens; Nova 2 Lite costs about US$0.30 and US$2.50 (AWS, n.d.-a).
Why we use it	Answers are generated only from retrieved, permitted chunks, so a small model is usually enough. Starting with Nova Lite keeps evaluation runs cheap; the team can switch to Nova 2 Lite if accuracy tests show a need.
How it helps the project	Supports the 40% accuracy criterion through grounded answers with source links, while thousands of test questions still cost only a few dollars.

3.9 AWS Systems Manager Parameter Store (standard tier)

Question	Answer
What it is	A free store for configuration values and secrets. Standard parameters carry no additional charge (AWS, n.d.-c) and can be encrypted with AWS KMS.
Why we use it	Tokens must never be stored in code or in the repository. Parameter Store gives encrypted storage at zero cost.
How it helps the project	Holds the Slack bot token, the Slack signing secret and the Sync Bot API token. Both Lambda functions read them at start-up and keep them in memory, so the number of calls stays very low (security criterion).

3.10 EventBridge Scheduler, CloudWatch and AWS Budgets

Question	Answer
What it is	EventBridge Scheduler starts the sync function on a fixed schedule (AWS, n.d.-l). CloudWatch stores logs and custom metrics. AWS Budgets sends an alert when spending passes a threshold (AWS, n.d.-m).
Why we use it	Freshness depends on a reliable schedule, and the team needs evidence (logs, freshness metric) for the evaluation. A budget alert is the last line of defence against unexpected charges.
How it helps the project	Supports the freshness and cost criteria and provides the audit log of who asked what and which pages were used.

D. Workflows and Sequence Diagrams

4.1 Provisioning test personas and signing in

Figure 2 shows how the System Admin (Group 7) prepares a test persona and how the persona signs in. In production, Katalon IT performs the first four steps through its own identity provider; in this environment the team performs them in Okta and Confluence.

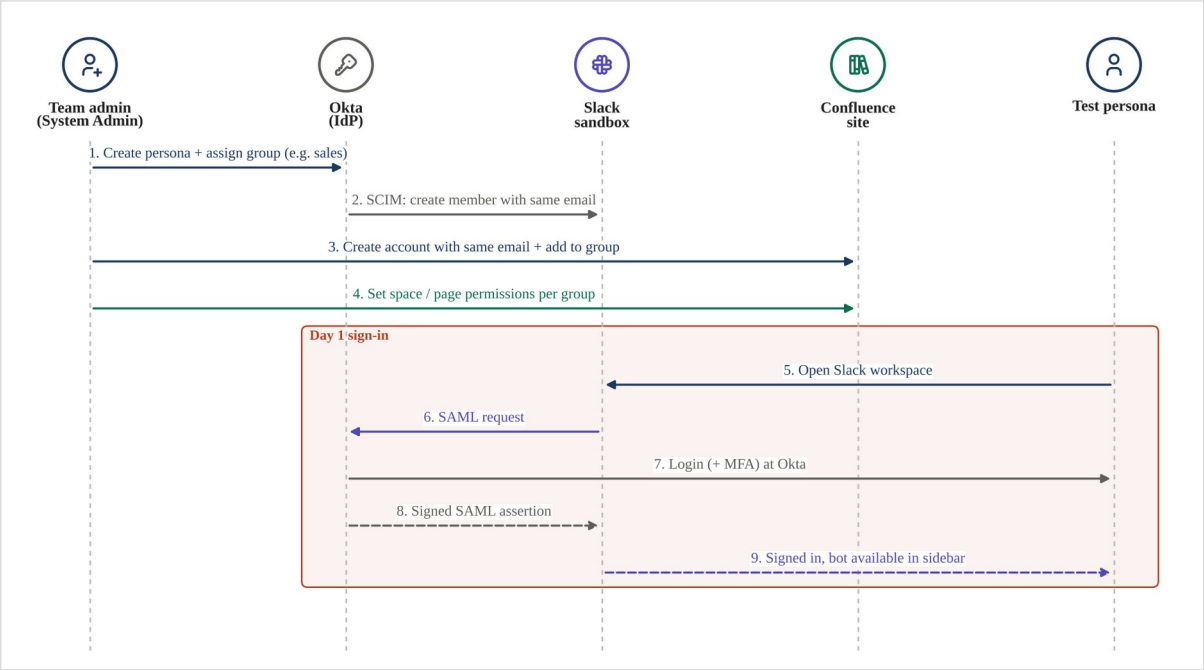


Figure 2. Sequence diagram: provisioning a persona and day-one sign-in

- **Steps 1–2:** the persona is created once in Okta, and SCIM creates the Slack member with the same email (Okta, n.d.).
- **Steps 3–4:** the Confluence account is created manually with the same email, because SCIM into Atlassian needs Atlassian Guard, which is not free for students. The account is added to a group, and the group decides which spaces and pages it can read.
- **Steps 5–9:** SAML single sign-on (Slack, n.d.-b). The bot never sees a password; it only receives the persona's Slack ID with each event.

4.2 Answering a question with two-layer access control

Figure 3 is the core workflow. The asker never signs in to the bot separately; identity comes from Slack, and access comes from Confluence.

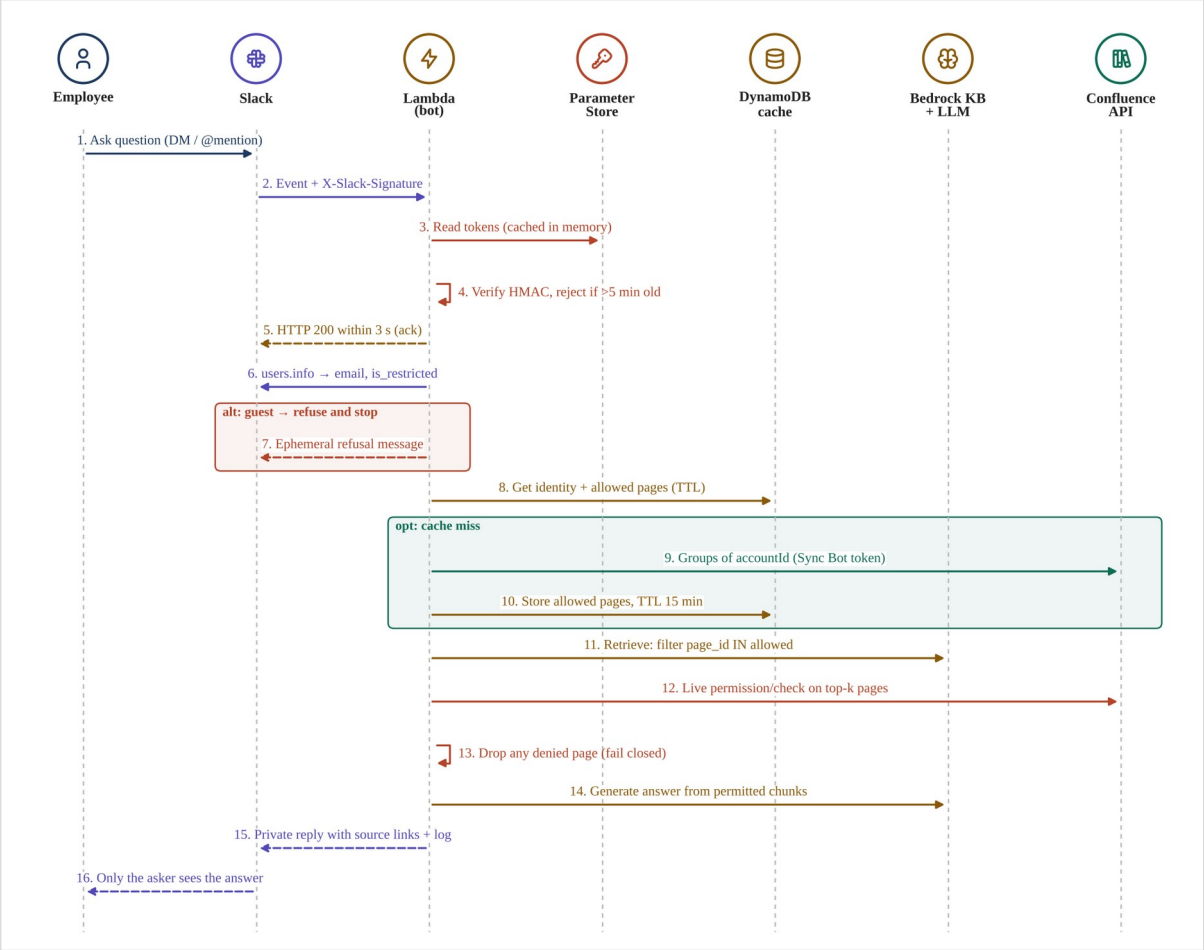


Figure 3. Sequence diagram: answering a question with a pre-filter and a live permission check

1. **Authenticate the request (steps 2–5).** The bot reads its tokens from Parameter Store, verifies the Slack signature and acknowledges within three seconds (Slack, n.d.-f, n.d.-g).
2. **Identify the asker (steps 6–7).** `users.info` returns the email and guest flags; guests receive a private refusal and processing stops (Slack, n.d.-c).
3. **Resolve permissions (steps 8–10).** The allowed page IDs come from the DynamoDB cache, or from Confluence on a cache miss, and are cached with a 15-minute TTL.
4. **Layer 1, pre-filter (step 11).** Bedrock `Retrieve` searches only chunks whose `page_id` is in the allowed list (AWS, n.d.-k).
5. **Layer 2, live check (steps 12–13).** The top results are checked again with the Confluence permission-check API (Atlassian, n.d.-d). Any page that is denied, or cannot be checked, is dropped: the system fails closed (NHI Mgmt Group, n.d.).
6. **Answer privately (steps 14–16).** The model writes the answer only from permitted chunks, and the reply is visible only to the asker, with source links. The question, user and page IDs are logged.

The second layer exists because a metadata filter is only as reliable as the identity and metadata passed to it. Recent RAG security research shows that checking identity only before retrieval is not sufficient, and recommends access decisions at both the retrieval and the generation stage (“Towards Secure Retrieval-Augmented Generation,” 2026; Yesil et al., 2026).

4.3 Keeping the knowledge base fresh

Figure 4 shows the incremental sync that serves the freshness criterion.

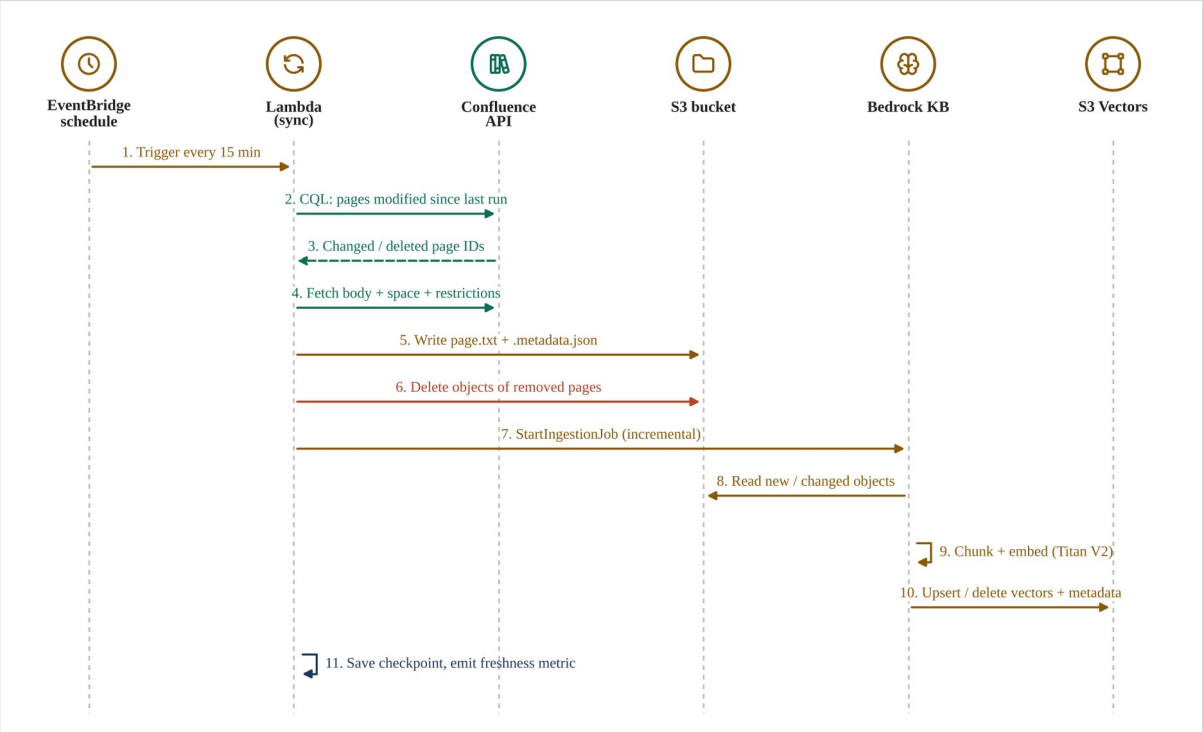


Figure 4. Sequence diagram: incremental synchronisation from Confluence to Bedrock

- **Change detection:** every 15 minutes the sync function asks Confluence, using CQL, for pages modified since the last checkpoint (Atlassian, n.d.-f).
- **Metadata:** each page is written to S3 with a `.metadata.json` file (page ID, space, groups), which is what makes the permission filter in Figure 3 possible (AWS, n.d.-k).
- **Incremental ingestion:** a Bedrock ingestion job processes only new, changed or deleted objects, so cost stays proportional to change (AWS, n.d.-n).
- **Evidence:** the function stores a checkpoint and publishes a freshness metric (minutes since the last successful sync) to CloudWatch for the evaluation.

E. Testing in This Environment

5.1 Test personas

Persona	Slack account	Confluence group	Expected result
Employee – Sales	Member	deal-desk	Answers only from Deal Desk and public Intranet pages
Employee – Partner	Member	partner-program	Answers only from Partner Program and public Intranet pages
Content Owner	Member	true-platform-admins (space admin)	Answers from True Platform; can force a re-sync
Client Reviewer	Member	reviewers	Same answers as an Employee; read-only dashboard
System Admin	Owner / Admin	none beyond own access	Manages the system but cannot read restricted pages
Guest	Single-channel guest	none	Refused with a private message

Table 2. Test personas (at most eight Slack users and two guests in the sandbox)

5.2 How access control is tested

1. **Build phase (Confluence Free):** the permission layer is tested with **mock permission data** and unit tests, including fail-closed behaviour when the Confluence check times out. No real permissions exist yet.
2. **Integration phase (Standard trial, 14 days):** permissions are configured in Confluence, the full persona matrix in Table 2 is run, and a page restriction is removed and re-added to measure how quickly the bot reacts.
3. **UAT:** the Client Reviewer runs the scenarios inside the same trial window. The trial must start at least one week before the planned UAT session.

F. Timeline

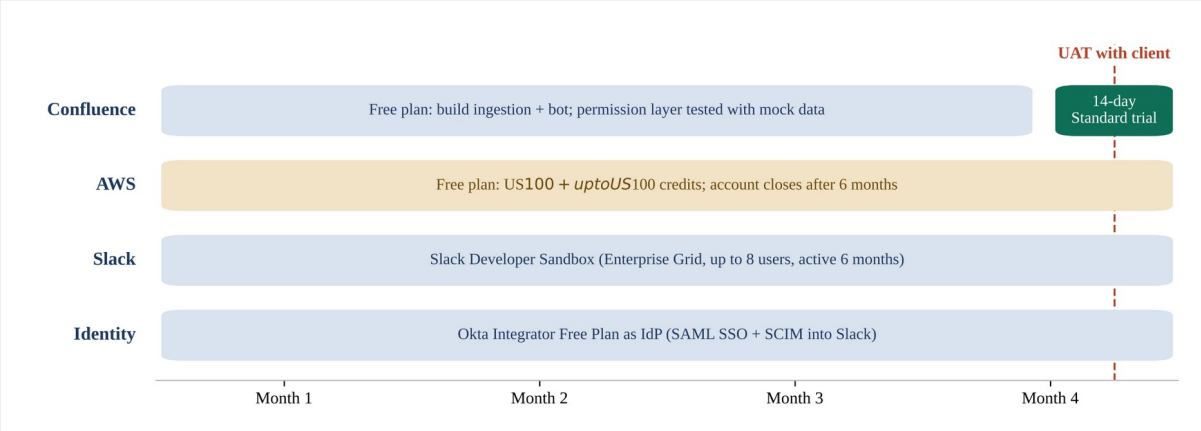


Figure 5. Four-month timeline of Option 1

The only time-critical element is the 14-day Standard trial. The AWS Free plan ends after six months or when the credits run out, whichever comes first (AWS, 2025a), so the AWS account should be opened when coding starts, not earlier.

G. Cost Summary

The detailed cost estimate, with labour hours and milestones, is in the matching budget workbook (Group 7, 2026b). The summary is:

Item	Estimate (4 months)	Paid by
Okta, Slack sandbox, Confluence Free + trial, Parameter Store	US$0	Free plans
Lambda, DynamoDB, EventBridge, CloudWatch	US$0	AWS always-free allowances
Bedrock (embeddings + Nova), S3 Vectors, S3	About US$7–15	AWS Free Tier credits (up to US$200)
Cash cost to the team	**US$0**	

New AWS customers receive US$100 in credits at sign-up and can earn another US$100 by using services such as Amazon EC2 and Amazon Bedrock (AWS, 2025a). On the Free plan no charge is made unless the account is upgraded (AWS, n.d.-e).

H. Risks and Mitigations

Risk	Effect	Mitigation
The 14-day trial is too short or badly timed	Access control is not fully tested before UAT	Prepare all test scripts in advance; start the trial one week before UAT; fall back to Option 2 for one month
Some services are unavailable on the AWS Free plan	Architecture must change late	On day one, create a Bedrock knowledge base on S3 Vectors as a smoke test
The AWS Free plan closes after six months	Environment disappears before the end	Open the account when coding starts; never join AWS Organizations, which upgrades the plan (AWS Builder Center, 2026)
Someone selects OpenSearch Serverless by mistake	About US$700 per month, credits gone in about a week	Use S3 Vectors only; AWS Budgets alert at US$1
Tokens leak through code or logs	Security incident	Store tokens only in Parameter Store; never log tokens; rotate the Slack token if exposed
Sandbox data differs from Katalon's real content	Accuracy results do not transfer	Build sample pages from the structure the client shared and ask the client to review them

I. Alignment with Evaluation Criteria

Criterion (weight)	How this environment supports it
Accuracy (40%)	Grounded answers from permitted chunks only, with source links; low model cost allows many evaluation runs.
Cost and maintainability (30%)	US$0 cash cost; S3 Vectors instead of OpenSearch Serverless; serverless services with no servers to maintain.
Security (20%)	Signature verification, guest refusal, two-layer permission check that fails closed, private replies, tokens in Parameter Store, audit log. Real permissions are tested during the trial window.
Creativity (10%)	A phased, zero-cost plan that still tests real access control, which is directly reusable by other student teams.

References

Amazon Web Services. (2024). *Access control for vector stores using metadata filtering with Knowledge Bases for Amazon Bedrock*. AWS Machine Learning Blog. https://aws.amazon.com/blogs/machine-learning/access-control-for-vector-stores-using-metadata-filtering-with-knowledge-bases-for-amazon-bedrock

Amazon Web Services. (2025a, July 16). *AWS Free Tier now offers $200 in credits and 6-month free plan to explore AWS at no cost*. What's New with AWS. https://aws.amazon.com/about-aws/whats-new/2025/07/aws-free-tier-credits-month-free-plan

Amazon Web Services. (2025b). *Amazon S3 Vectors now generally available with increased scale and performance*. AWS News Blog. https://aws.amazon.com/blogs/aws/amazon-s3-vectors-now-generally-available-with-increased-scale-and-performance

Amazon Web Services. (n.d.-a). *Amazon Bedrock pricing*. Retrieved November 11, 2026, from https://aws.amazon.com/bedrock/pricing/

Amazon Web Services. (n.d.-b). *AWS Secrets Manager pricing*. Retrieved November 11, 2026, from https://aws.amazon.com/secrets-manager/pricing/

Amazon Web Services. (n.d.-c). *AWS Systems Manager pricing*. Retrieved November 11, 2026, from https://aws.amazon.com/systems-manager/pricing/

Amazon Web Services. (n.d.-d). *Amazon S3 pricing*. Retrieved November 11, 2026, from https://aws.amazon.com/s3/pricing/

Amazon Web Services. (n.d.-e). *AWS Free Tier*. Retrieved November 11, 2026, from https://aws.amazon.com/free/

Amazon Web Services. (n.d.-f). *Creating and managing Lambda function URLs*. AWS Lambda Developer Guide. Retrieved November 11, 2026, from https://docs.aws.amazon.com/lambda/latest/dg/urls-configuration.html

Amazon Web Services. (n.d.-g). *AWS Lambda pricing*. Retrieved November 11, 2026, from https://aws.amazon.com/lambda/pricing/

Amazon Web Services. (n.d.-h). *Using time to live (TTL) in DynamoDB*. Amazon DynamoDB Developer Guide. Retrieved November 11, 2026, from https://docs.aws.amazon.com/amazondynamodb/latest/developerguide/TTL.html

Amazon Web Services. (n.d.-i). *Amazon DynamoDB pricing*. Retrieved November 11, 2026, from https://aws.amazon.com/dynamodb/pricing/

Amazon Web Services. (n.d.-j). *Amazon Titan Text Embeddings models*. Amazon Bedrock User Guide. Retrieved November 11, 2026, from https://docs.aws.amazon.com/bedrock/latest/userguide/titan-embedding-models.html

Amazon Web Services. (n.d.-k). *Connect to Amazon S3 for your knowledge base*. Amazon Bedrock User Guide. Retrieved November 11, 2026, from https://docs.aws.amazon.com/bedrock/latest/userguide/s3-data-source-connector.html

Amazon Web Services. (n.d.-l). *What is Amazon EventBridge Scheduler?* Retrieved November 11, 2026, from https://docs.aws.amazon.com/scheduler/latest/UserGuide/what-is-scheduler.html

Amazon Web Services. (n.d.-m). *AWS Budgets*. Retrieved November 11, 2026, from https://aws.amazon.com/aws-cost-management/aws-budgets/

Amazon Web Services. (n.d.-n). *Sync your data with your Amazon Bedrock knowledge base*. Amazon Bedrock User Guide. Retrieved November 11, 2026, from https://docs.aws.amazon.com/bedrock/latest/userguide/kb-data-source-sync-ingest.html

amitrix. (n.d.). *Which tier does your vector workload live on?* DEV Community. Retrieved November 11, 2026, from https://dev.to/amitrix/which-tier-does-your-vector-workload-live-on-5f2a

Atlassian. (n.d.-a). *Purchasing & licensing FAQ*. Retrieved November 11, 2026, from https://www.atlassian.com/licensing

Atlassian. (n.d.-b). *Atlassian Classroom license request*. Retrieved November 11, 2026, from https://www.atlassian.com/solutions/survey/classroom-license-request

Atlassian. (n.d.-c). *Confluence pricing*. Retrieved November 11, 2026, from https://www.atlassian.com/software/confluence/pricing

Atlassian. (n.d.-d). *The Confluence Cloud REST API: Content permissions*. Atlassian Developer. Retrieved November 11, 2026, from https://developer.atlassian.com/cloud/confluence/rest/v1/api-group-content-permissions/

Atlassian. (n.d.-e). *Manage API tokens for your Atlassian account*. Atlassian Support. Retrieved November 11, 2026, from https://support.atlassian.com/atlassian-account/docs/manage-api-tokens-for-your-atlassian-account/

Atlassian. (n.d.-f). *Advanced searching using CQL*. Atlassian Developer. Retrieved November 11, 2026, from https://developer.atlassian.com/cloud/confluence/advanced-searching-using-cql/

AWS Builder Center. (2026). *The $0 AWS account in 2026: Free plan, credits, guardrails*. Retrieved November 11, 2026, from https://builder.aws.com/content/3Kkt4kNiDSUnO3n6db2weSNubzE/the-dollar0-aws-account-in-2026-free-plan-credits-guardrails

Group 7. (2026a). *Environment options* (Version 1.0, draft) [Unpublished project document]. COS40005 Computing Technology Project A, Swinburne University of Technology.

Group 7. (2026b). *Project budget: Option 1, zero cost* (Version 1.0, draft) [Unpublished spreadsheet]. COS40005 Computing Technology Project A, Swinburne University of Technology.

Mallesh, S. (n.d.). *S3 Vectors for RAG: The cheapest vector store for Bedrock Knowledge Bases with Terraform*. DEV Community. Retrieved November 11, 2026, from https://dev.to/suhas_mallesh/s3-vectors-for-rag-the-cheapest-vector-store-for-bedrock-knowledge-bases-with-terraform-487c

NHI Mgmt Group. (n.d.). *Permission-aware RAG guide*. Retrieved November 11, 2026, from https://nhimg.org/permission-aware-rag-guide

Okta. (n.d.). *Create an app integration*. Okta Developer. Retrieved November 11, 2026, from https://developer.okta.com/docs/guides/build-sso-integration/saml2/test-your-app/

Slack. (n.d.-a). *Developer sandboxes*. Slack Developer Docs. Retrieved November 11, 2026, from https://docs.slack.dev/tools/developer-sandboxes/

Slack. (n.d.-b). *Set up SAML single sign-on for Slack*. Slack Help Center. Retrieved November 11, 2026, from https://slack.com/help/articles/203772216

Slack. (n.d.-c). *users.info method*. Slack API. Retrieved November 11, 2026, from https://api.slack.com/methods/users.info

Slack. (n.d.-d). *users:read.email scope*. Slack API. Retrieved November 11, 2026, from https://api.slack.com/scopes/users:read.email

Slack. (n.d.-e). *User object*. Slack Developer Docs. Retrieved November 11, 2026, from https://docs.slack.dev/reference/objects/user-object

Slack. (n.d.-f). *Verifying requests from Slack*. Slack API. Retrieved November 11, 2026, from https://api.slack.com/authentication/verifying-requests-from-slack

Slack. (n.d.-g). *Events API*. Slack API. Retrieved November 11, 2026, from https://api.slack.com/apis/events-api

Slack. (n.d.-h). *Testing Enterprise Grid apps*. Slack API. Retrieved November 11, 2026, from https://api.slack.com/enterprise/testing

Towards secure retrieval-augmented generation: A comprehensive review of threats, defenses and benchmarks (arXiv:2603.21654). (2026). arXiv. https://arxiv.org/pdf/2603.21654

Yesil, H., Gultekin, S., Bozyigit, F., Tezel, B. T., & Challenger, M. (2026). *A database-independent LLM framework for real-time authorization in retrieval-augmented generation* [Workshop paper]. AAAI 2026 Workshop on Trustworthy Agentic AI. https://trustagenticai.github.io/AAAI2026/AAAI-Workshop/68.pdf